

Dr. Swaetha Ramkumar

Postdoctoral Research Scholar | Arizona State University

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Education

- 2021 – 2025  **Ph.D. in Astrophysics**, Trinity College Dublin
Thesis title: *Probing Ultra-Hot Jupiters with CRIRES+: Atmospheric retrievals from phase-resolved emission spectroscopy*
Supervisor: Prof. Neale Gibson
- 2019 – 2020  **M.Sc. in Astrophysics**, University College London (UCL)
Supervisor: Dr. Ralph Schoenrich
Distinction
- 2016 – 2019  **B.Sc. in Physics**, Amrita Vishwa Vidyapeetham
Supervisor: Dr. Bharat Kishore Sharma
First Class with Distinction (CGPA: 9.10 out of 10)

Research Experience

- Feb 2026 – Present  **Postdoctoral Research Scholar**, Arizona State University
- Contributing to the MAPS (MMT Adaptive optics exoPlanet characterization System) program — an instrument development and observational science initiative at the 6.5m MMT Observatory encompassing adaptive optics, and atmospheric characterisation of exoplanets; including development of the data reduction pipeline for the upgraded ARIES spectrograph.
 - Developing atmospheric retrieval frameworks for high-resolution spectroscopic data from ground-based facilities.
- Sept 2021 – Sept 2025  **PhD Researcher**, Trinity College Dublin
- Atmospheric characterisation of exoplanets using high-resolution emission spectroscopy.
 - Applied advanced Bayesian inference techniques to constrain atmospheric composition, T-P profiles, and dynamics of ultra-hot Jupiters.
 - Analysed day-side atmospheres across different phase sequences with VLT/CRIRES+ to investigate variations in atmospheric properties as the planet rotates.
- Mar 2020 – Sep 2020  **Master's Research Project**, University College London (UCL)
- Studied the non-axisymmetric bar of the Milky Way and performed simulations using an orbit integrator (written in C++).
 - Investigated the behaviour of x_2 orbits and their interactions with the bar.
 - Explored the behaviour of orbital resonances and x_2 orbits when introducing a nuclear disk.

Research Experience (continued)

Feb 2019 – May 2019

■ **Undergraduate Research Project**, Amrita Vishwa Vidyapeetham

- Investigated the basic properties of Solar and Interstellar Plasma.
- The simulation output was investigated in Python to determine the plasma parameters (such as Debye length and Debye number) as a function of temperature. These results were then compared with observed values in the Solar wind and Interstellar medium.

Research Publications

First-Authored

- 1 **S. Ramkumar**, Gibson, Neale P., Nugroho, Stevanus K., Fortune, Mark, and Maguire, Cathal, “New perspectives on mascara-1b: A combined analysis of pre- and post-eclipse emission data using crires+,” *A&A*, vol. 695, A110, 2025.  DOI: [10.1051/0004-6361/202453520](https://doi.org/10.1051/0004-6361/202453520).
- 2 **S. Ramkumar**, N. P. Gibson, S. K. Nugroho, C. Maguire, and M. Fortune, “High-resolution emission spectroscopy retrievals of MASCARA-1b with CRIRES+: strong detections of CO, H₂O, and Fe emission lines and a C/O consistent with solar,” *MNRAS*, 2023.  DOI: [10.1093/mnras/stad2476](https://doi.org/10.1093/mnras/stad2476).

Co-Authored

- 1 M. Fortune, N. P. Gibson, D. Foreman-Mackey, T. M. Evans-Soma, C. Maguire, and **S. Ramkumar**, “How do wavelength correlations affect transmission spectra? Application of a new fast and flexible 2D Gaussian process framework to transiting exoplanet spectroscopy,” *A&A*, 2024.  DOI: [10.1051/0004-6361/202347613](https://doi.org/10.1051/0004-6361/202347613).
- 2 C. Maguire, N. P. Gibson, S. K. Nugroho, M. Fortune, **S. Ramkumar**, S. Gandhi, and E. de Mooij, “High resolution atmospheric retrievals of WASP-76b transmission spectroscopy with ESPRESSO: Monitoring limb asymmetries across multiple transits,” *A&A*, 2024.  DOI: [10.1051/0004-6361/202449449](https://doi.org/10.1051/0004-6361/202449449).
- 3 C. Maguire, N. P. Gibson, S. K. Nugroho, **S. Ramkumar**, M. Fortune, S. R. Merritt, and E. de Mooij, “High-resolution atmospheric retrievals of WASP-121b transmission spectroscopy with ESPRESSO: Consistent relative abundance constraints across multiple epochs and instruments,” *MNRAS*, 2023.  DOI: [10.1093/mnras/stac3388](https://doi.org/10.1093/mnras/stac3388).

Talks and Presentations

- 2025
- **New perspectives on MASCARA-1b: Probing pre- and post-eclipse emission with CRIRES+**
ExoClimes VII, July 7-11, 2025 (poster presentation).
 - **The day-side atmosphere of MASCARA-1b through the eyes of CRIRES+**
EAS Annual Meeting, June 23-27, 2025 (poster presentation).
 - **Probing Ultra-Hot Jupiter Atmospheres with Phase-Resolved Emission Spectroscopy**
Trinity College Dublin Postgraduate Seminar, May 2025 (seminar talk).
 - **New perspectives on MASCARA-1b**
Trinity College Dublin Astrophysics Seminar, May 2025 (seminar talk).
 - **The Cosmic Blowtorch: Planets Under Extreme Heat**
Three Minute Thesis (3MT), Trinity College Dublin – Heats & Final (March, 2025)..
 - **“Ultra-hot” Jupiters: Where a Year Lasts a Day**
IOP Ireland Spring Conference: Rosse Medal entrant, Feb 28-01 Mar 2025 (poster presentation).

Talks and Presentations (continued)

- 2024  **MASCARA: Does it help your eyelash?**
Two HoRSEs, July 15-19, 2024 (poster presentation).
-  **MASCARA: Does it help your eyelash?**
Exoplanets 5, June 17-21, 2024 (poster presentation).
-  **Atmospheres of Alien Worlds.**
IOP Ireland Spring Conference: Rosse Medal entrant, Apr 06, 2024 (poster presentation).
- 2023  **MASCARA: does it help your eyelash?**
Irish National Astronomy Meeting (INAM) 2023, Aug 24-25, 2023 (contributed talk).
-  **High-resolution emission spectroscopy retrievals of MASCARA-1b with CRIRES+**
Exoplanets by the Lake Summer School, Jul 31-Aug 4, 2023 (contributed talk).
-  **High-resolution emission spectroscopy retrievals of MASCARA-1b with CRIRES+**
Trinity College Dublin Astrophysics Seminar (seminar talk).
-  **High-resolution emission spectroscopy retrievals of MASCARA-1b with CRIRES+**
2023 Sagan Exoplanet Summer Hybrid Workshop, Jul 24-28, 2023 (poster presentation).
-  **The atmosphere of MASCARA-1b through the eyes of CRIRES+**
Theo Murphy meeting, the Royal Society: Spectroscopy of exoplanets at high resolution, Feb 6-7, 2023 (flash talk).

Observing Experience and Proposals

- 2026  **MIRAC-5 and ARIES at the MMT Observatory**
Commissioning observations over 6 nights to characterise AO performance, and test instrument functionality in preparation for science operations.
- 2024  **CRIRES+ at the Very Large Telescope (VLT)**
Phase Curve observations (K-band) during cycle P113, PI: Nugroho, CoI: S. Ramkumar.
- 2023  **CRIRES+ at the Very Large Telescope (VLT)**
Phase Curve observations (K-band) during cycle P112, PI: Gibson, dPI: S. Ramkumar.

Teaching and Outreach

- June 2025  **Session Chair**
EAS Annual Meeting, Cork
Symposium 14: New Frontiers in Characterising Gas-Giant Exoplanets and Brown Dwarfs
-  **Scientific Organiser**
EAS Annual Meeting, Cork
Symposium 14: New Frontiers in Characterising Gas-Giant Exoplanets and Brown Dwarfs
- 2021 – 2025  **Teaching Assistant in PYU33AP4 - JS AP Astro Computational Lab**
Trinity College Dublin
- Oct 2023 – Nov 2023  **Teaching Assistant in PYU33AP3 - JS Practical in Astrophysics**
Trinity College Dublin
- Apr 2023  **Transition Year Physics Experience (TYPE) Mentor**
Trinity College Dublin

Teaching and Outreach (continued)

Nov 2022 – Mar 2023  **STEM@Universi-TY Educator**
Trinity Walton Club, Trinity College Dublin
<https://www.tcd.ie/waltonclub/ty.php>

Prizes, Awards & Grants

- 2021 – 2025  **Research Grant**, Provost's PhD Award
Trinity College Dublin
Recipient of a full scholarship to undertake doctoral-level research at Trinity for four years.
- March 2025  **Three Minute Thesis (3MT) Finalist**
Trinity College Dublin
Finalist in the university-wide 3MT competition, presenting PhD research in three minutes using a single slide, to a non-specialist audience
[3MT Slide & Heat Photos](#).
- June 2024  **Science in Shorts 2024**
Nature Awards
Science in Shorts is one of Nature's Awards, where you present your research in a 1-minute video. My video was selected for inclusion in the Shortlist and is featured on their YouTube channel:
[Science in Shorts: Turn into a force ghost!](#)
- Aug 2023  **Peter Curran Award**
Astronomical Society of Ireland (ASI)
Best student talk at the Irish National Astronomy Meeting (INAM) for the year 2023.
<https://astronomers.ie/peter-curran-award/>

Technical Skills

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| Research Interests |  Exoplanet atmospheres (observations and modelling), Low- and High-resolution spectroscopy, Cross-correlation analysis, Atmospheric retrievals, Planet formation. |
| Programming |  Python, C/C++ (<i>intermediate</i>), SQL (<i>basic</i>) |
| Markup Languages |  $\text{\LaTeX}$, HTML/CSS |
| Design & Publishing |  Affinity Designer, Affinity Publisher |
| Data Visualisation |  Matplotlib, Gnuplot, Seaborn |
| Miscellaneous |  Bayesian inference with MCMC, Cross-correlation analysis, Web development, Data Reduction pipelines |

Languages

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| English |  Full professional proficiency |
| Tamil |  Native or bilingual proficiency |
| Hindi |  Limited working proficiency |